

PA4 - SUMMATIVE USER STUDY REPORT

Course: Interface Design (CSC13112)

Instructors: Dr. Le Khanh Duy, MSc. Pham Nguyen Son Tung

Group: Group 06 - **Class:** 23KTMP2

Team members:

- Le Mai Hoai Bao (23127326)
 - Lam Huu Khanh (23127205)
 - Pham Chi Bao Ninh (23127446)
 - Truong Cong Thien Phu (23127455)
 - Phung Ngoc Tuan (23127510)
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1. Research Objectives & Experimental Hypotheses

1.1. Context & Evaluation Objectives

After completing formative testing on paper prototypes in PA3 (Formative Testing with 6 variants), the team selected the optimal architecture:

- **Navigation system:** A 5-tab fixed bottom navigation bar (**Nav-1: Fixed Bottom Navigation Bar**).
- **Schedule screen:** A horizontal date strip combined with match cards and an 8×8 chessboard detail view (**Sch-2: Date Strip Filter + Match Detail**).

This Summative Usability Study was conducted to **quantify and validate the superiority** of the high-fidelity prototype (**Group 06 Hi-fi Prototype**) compared with the current mobile version of the original product (**Baseline: freestylechess.com mobile web**).

1.2. Formal Hypotheses

The team proposed 4 controlled scientific hypotheses based on the 4 core improvements from PA3:

- **Hypothesis H_1 (Time-on-Task / Efficiency):** The average task completion time (TCT) on the Hi-fi Prototype will decrease by at least **40%** compared with the Baseline across both tasks (Task 1: one-handed navigation; Task 2: looking up the schedule and viewing chessboard details).

$$\mu_{TCT(Prototype)} < \mu_{TCT(Baseline)}$$

- **Hypothesis H_2 (Overall Usability / SUS Benchmark):** The **System Usability Scale (SUS Score)** of the Hi-fi Prototype will exceed the industry benchmark ($\geq 68/100$, reaching B+ to A according to the Sauro-Lewis scale) and be at least **20 points** higher than the Baseline.

$$SUS_{Prototype} \geq 80 > SUS_{Baseline}$$

- **Hypothesis H_3 (Effectiveness & Error Reduction):** The Hi-fi Prototype will achieve a **Direct Success Rate** of $\geq 90\%$, while reducing slip/touch-target misses and mental model mistakes by complying with the thumb touch-target size standard of $\geq 48 \times 48$ px.

$$Errors_{Prototype} \ll Errors_{Baseline}$$

- **Hypothesis H_4 (Subjective Satisfaction):** The 5-point Likert scale scores across all 5 dimensions (*Ease of use, Aesthetics, Speed, Navigation clarity, Overall satisfaction*) for the Prototype will average $\geq 4.2/5.0$, statistically

outperforming the Baseline.

2. Experimental Study Design

2.1. Within-Subjects Design

- **Model:** Within-Subjects / Repeated Measures Design.
- **Rationale:**
 - Completely eliminates variation in individual ability among participants (inter-individual differences) for a small sample size ($N = 6$).
 - Allows participants to visually and directly compare the 2 systems during the semi-structured interview at the end of the session.
- **Independent Variable (IV):** The interface system being tested:
 - **System A (Baseline):** The actual mobile website freestylechess.com.
 - **System B (Hi-fi Prototype):** The interactive web prototype running in a mobile browser, developed by Group 06.
- **Dependent Variables (DV):**
 - Task Completion Time (seconds).
 - Task Success Rate: Direct / Partial / Fail.
 - Error Frequency, classified by Slip, Mistake, Hesitation.
 - SUS usability score (0 - 100).
 - Likert satisfaction score (1 - 5 across 5 dimensions).
 - Qualitative feedback and user preference.

2.2. Extraneous Variable Control & Counterbalancing Matrix

To reduce the **Learning / Carryover Effect**, where users remember information locations from the first system and apply them to the second system, the task order was cross-assigned using two orders, AB/BA . The actual allocation was **4 participants using A → B and 2 participants using B → A**; therefore, this is partial counterbalancing rather than perfect balancing.

Participant	Demographics & Habits	Video Evidence (YouTube)	Testing order (Order)	System 1 (First)	System 2 (Second)
P01	Right-handed, does not follow chess news, often uses one hand	Video P01	A → B	Baseline (freestylechess.com)	Hi-fi Prototype
P02	Right-handed, plays chess frequently (ELO ~1400), passively follows chess	Video P02	A → B	Baseline (freestylechess.com)	Hi-fi Prototype
P03	Right-handed, not interested in chess, late 20s	Video P03	A → B	Baseline (freestylechess.com)	Hi-fi Prototype
P04	Right-handed, often uses two	Video P04	A → B	Baseline (freestylechess.com)	Hi-fi Prototype

Participant	Demographics & Habits	Video Evidence (YouTube)	Testing order (Order)	System 1 (First)	System 2 (Second)
	hands, not interested in chess				
P05	Right-handed, uses one hand when browsing media / two hands when gaming	Video P05	B → A	Hi-fi Prototype	Baseline (freestylechess.com)
P06	Right-handed, knows how to play chess, often uses one hand when watching videos	Video P06	B → A	Hi-fi Prototype	Baseline (freestylechess.com)

3. Standardized Study Protocol

Each testing session lasted about **20 minutes** and followed 9 steps:

- Step 1: Welcome, Introduction & Consent Agreement
- Step 2: Collect Demographic Information Form
- Step 3: Perform Scenario on SYSTEM 1 (Task 1 & Task 2)
- Step 4: Complete SYSTEM 1 Survey (SUS + Likert Satisfaction)
- Step 5: Cognitive wash-out period (90s)
- Step 6: Perform Scenario on SYSTEM 2 (Task 1 & Task 2)
- Step 7: Complete SYSTEM 2 Survey (SUS + Likert Satisfaction)
- Step 8: Comparative Semi-Structured Post-Study Interview
- Step 9: Debriefing, Thanks, Souvenir Gift & End Session

4. Task Specifications & Assistance Protocol

Task 1: Single-Handed Navigation & Ergonomics

- **Realistic scenario:** "You are walking on the street and have only one hand free to hold your phone. You want to quickly check the upcoming schedule for the Freestyle Chess tournament."
- **Task prompt:** "From the current home screen, use **ONLY** one thumb to find and open the Schedule page."
- **Timing points (TCT_1):**
 - **Start (t_{start}):** As soon as the facilitator says "Start".
 - **End (t_{end}):** When the Schedule screen is fully displayed and the user stops tapping.
- **Success criteria:**
 - **Direct Success:** Accurately taps the **Schedule** tab on the Bottom Nav (Prototype) or finds the Schedule item on the Baseline within ≤ 60 s without any assistance.
 - **Partial/Assisted Success:** Completes the task after receiving 01 standardized hint (after being stuck for more than 45 s).
 - **Fail/Timeout:** Exceeds 120 s or the participant states that they give up.

Task 2: Schedule Lookup & 8×8 Chessboard Match Detail

- **Realistic scenario:** "You know that on 02/09/2026 there is a match between Magnus Carlsen and Le Quang Liem. You want to view detailed information about this matchup and see the 8×8 chessboard position layout."
- **Task prompt:** "On the Schedule page, find the match on 02/09/2026 between Magnus Carlsen and Le Quang Liem, then open the match detail screen to view the analysis chessboard."
- **Timing points (TCT_2):**
 - **Start (t_{start}):** As soon as the facilitator says "Start".
 - **End (t_{end}):** When the Match Detail route (`/schedule/:matchId`) opens with the 8×8 chessboard and information for both players clearly displayed.
- **Success criteria:**
 - **Direct Success:** Interacts with the date strip to select 02/09/2026 → expands the match card → taps **Chi tiết / Detail** within ≤ 60 s.
 - **Partial Success:** Completes the task after 01 standardized hint when stuck for more than 60 s.
 - **Fail/Timeout:** Exceeds 120 s or cannot find the match.

Standardized Assistance Protocol

To ensure objectivity and reproducibility, the facilitator may intervene only at 2 fixed levels:

1. **Level 1 Hint (Only when the user is silent/stuck for > 45 s in Task 1 or > 60 s in Task 2):** "You may pay attention to the toolbars at the bottom or the date filter area at the top of the screen."
2. **Stop task (When reaching the 120 s timeout threshold):** "Thank you, we can stop this task here and continue to the next step."

5. HCI Usability Breakdown & Error Taxonomy

The observer will use the following 3-category behavioral error coding table to record issues in the log:

Error code	Error Category	Definition & Observable Manifestation	Root Cause
ERR-1	Slip / Touch Target Miss	Misses the target, accidentally taps an adjacent tab, or the thumb cannot reach a touch point near the top.	Button size $< 48\text{ px}$, <i>spacing between buttons</i> $< 8\text{ px}$, outside the ergonomic Thumb Reach Zone.
ERR-2	Mental Model Mistake	Clicks unrelated menus (such as News, Videos, Settings) while looking for Schedule; does not know how to expand the card.	Lack of visual affordance; messy hierarchical information architecture (IA).
ERR-3	Hesitation Delay (cognitive hesitation > 3 s)	Finger pauses above the screen, eyes scan back and forth continuously for $> 3\text{ s}$ before tapping; doubts whether a button is tappable.	Lack of visual feedback, active state, or clear loading feedback.

6. Measurement Instruments

6.1. System Usability Scale - SUS

The study applied John Brooke's (1996) international standard scale consisting of 10 bilingual questions with a 5-point Likert scale (1 = Strongly disagree, 5 = Strongly agree).

Standardized SUS scoring formula:

- For **odd-numbered** questions (1, 3, 5, 7, 9 - positive statements):

$$Score_i = X_i - 1$$

- For **even-numbered** questions (2, 4, 6, 8, 10 - negative statements):

$$Score_i = 5 - X_i$$

- Converted score on a 100-point scale: **SUS = (Score₁ + Score₂ + ... + Score₁₀) × 2.5**

SUS Score Rating Scale (Sauro & Lewis, Bangor et al.):

- SUS ≥ 80.3:** Grade **A / A+** (Excellent — Superior experience).
- 68.0 ≤ SUS < 80.3:** Grade **B / C** (Good / Above Average — Industry benchmark).
- 51.0 ≤ SUS < 68.0:** Grade **D** (Poor / OK — Needs many improvements).
- SUS < 51.0:** Grade **F** (Unacceptable — Failed experience).

6.2. 5-Dimension Likert Satisfaction Scale

Evaluates agreement level (1 to 5) across 5 dimensions:

- Q₁ - Ease of Use:** "I find it very easy to search for and view chess match information in this application."
- Q₂ - Aesthetics:** "The application's interface looks beautiful, professional, modern, and appealing."
- Q₃ - Speed & Responsiveness:** "The application loads quickly and responds immediately to my touch interactions."
- Q₄ - Navigation Clarity:** "The buttons, navigation bar, and options are intuitive, easy to find, and easy to understand."
- Q₅ - Overall Satisfaction:** "Overall, I am very satisfied with the experience of using this application."

7. Raw Data Logging Sheets**7.1. Quantitative Performance Log****A. Baseline System (freestylechess.com mobile web)**

Participant	Order	Task 1: TCT (s)	Task 1: Result	Task 2: TCT (s)	Task 2: Result	ERR-1 (Slip)	ERR-2 (Mistake)	ERR-3 (Hesitation)	SUS Score (0-100)	Likert Avg (1-5)
P01	1st	38.2	Direct	84.5	Partial (1 hint)	4	2	3	27.5	2.8
P02	1st	29.4	Direct	62.1	Direct	2	1	2	50.0	3.4
P03	1st	41.0	Partial (1 hint)	92.0	Partial (1 hint)	5	3	4	22.5	2.6
P04	1st	28.5	Direct	58.4	Direct	2	1	2	50.0	3.2

Participant	Order	Task 1: TCT (s)	Task 1: Result	Task 2: TCT (s)	Task 2: Result	ERR-1 (Slip)	ERR-2 (Mistake)	ERR-3 (Hesitation)	SUS Score (0-100)	Likert Avg (1-5)
P05	2nd	36.0	Direct	77.0	Direct	3	2	3	37.5	3.0
P06	2nd	44.5	Partial (1 hint)	120.0+	Fail / Assist	5	4	5	12.5	2.4
Average ($\mu \pm \sigma$)	—	36.3 $\pm$ 6.4s	67% Direct / 33% Part	82.3 $\pm$ 22.8s	50% Direct / 50% Part-Fail	3.5 $\pm$ 1.4	2.2 $\pm$ 1.1	3.2 $\pm$ 1.1	33.3 $\pm$ 15.2 (Grade F)	2.9 $\pm$ 0.4

B. Hi-fi Prototype System (Group 06)

Participant	Order	Task 1: TCT (s)	Task 1: Result	Task 2: TCT (s)	Task 2: Result	ERR-1 (Slip)	ERR-2 (Mistake)	ERR-3 (Hesitation)	SUS Score (0-100)	Likert Avg (1-5)
P01	2nd	11.2	Direct	14.8	Direct	0	0	1	97.5	4.8
P02	2nd	10.5	Direct	13.9	Direct	0	0	0	92.5	4.6
P03	2nd	14.1	Direct	18.2	Direct	0	1	1	75.0	4.2
P04	2nd	11.0	Direct	13.5	Direct	0	0	0	97.5	4.8
P05	1st	12.2	Direct	15.6	Direct	0	0	1	92.5	4.6
P06	1st	12.0	Direct	12.0	Direct	0	0	0	100.0	5.0
Average ($\mu \pm \sigma$)	—	11.8 $\pm$ 1.2s	100% Direct	14.7 $\pm$ 2.1s	100% Direct	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	0.5 $\pm$ 0.5	92.5 $\pm$ 9.1 (Grade A+)	4.7 $\pm$ 0.3

7.2. SUS Item Breakdown Matrix

System	P_ID	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total SUS Score	Grade
Baseline	P01	2	4	2	4	2	4	3	4	2	4	27.5	Grade F
	P02	3	3	3	3	3	3	3	3	3	3	50.0	Grade F
	P03	2	4	2	5	2	4	2	4	2	4	22.5	Grade F
	P04	3	3	3	3	3	3	3	3	3	3	50.0	Grade F
	P05	3	4	3	4	3	3	3	4	2	4	37.5	Grade F
	P06	2	5	2	5	2	5	2	5	2	5	12.5	Grade F

System	P_ID	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total SUS Score	Grade
Prototype	P01	5	1	5	1	4	1	5	1	5	1	97.5	Grade A+
	P02	5	1	4	2	5	1	5	1	4	1	92.5	Grade A+
	P03	4	2	4	2	4	2	4	2	4	2	75.0	Grade B
	P04	5	1	5	1	4	1	5	1	5	1	97.5	Grade A+
	P05	5	1	4	2	4	1	5	1	5	1	92.5	Grade A+
	P06	5	1	5	1	5	1	5	1	5	1	100.0	Grade A+

7.3. 5-Dimension Likert Scale Detailed Summary

System	P_ID	Q1: Ease	Q2: Aesthetics	Q3: Speed	Q4: Nav Clarity	Q5: Satisfaction	Likert Avg
Baseline	P01	3	3	3	2	3	2.8
	P02	3	4	3	3	4	3.4
	P03	2	3	3	2	3	2.6
	P04	3	4	3	3	3	3.2
	P05	3	3	3	3	3	3.0
	P06	2	3	2	2	3	2.4
Prototype	P01	5	5	5	5	4	4.8
	P02	5	5	4	5	4	4.6
	P03	4	4	4	5	4	4.2
	P04	5	5	5	5	4	4.8
	P05	5	5	4	5	4	4.6
	P06	5	5	5	5	5	5.0

8. Data Synthesis & Visualization Plan

8.1. Task Completion Time Analysis

- **Task 1 (one-handed navigation):** TCT decreased from 36.3 s to 11.8 s (a 67.5% reduction, $p < 0.001$).
- **Task 2 (schedule lookup & chessboard):** TCT decreased from 82.3 s to 14.7 s (an 82.1% reduction, $p < 0.001$). In particular, P06 spent more than 120 s on the old system due to being led into a news article, but took exactly 12 s on the Prototype.

8.2. Error Reduction Analysis

- **Slip Errors (ERR-1):** Decreased from an average of 3.5 errors to 0.0 errors thanks to touch targets of $\geq 48 \times 48$ px.
- **Mental Model Mistakes (ERR-2):** Decreased from 2.2 to 0.2 errors thanks to the fixed navigation bar and clear card hierarchy.
- **Hesitation Delays (ERR-3):** Decreased from 3.2 to 0.5 times thanks to instant visual feedback < 250 ms.

8.3. SUS & Likert Scale Comparison

- **SUS Benchmark:** Increased from 33.3/100 (**Grade F**) to 92.5/100 (**Grade A+**), a difference of +59.2 points, exceeding the industry benchmark of 68.0.
- **Likert Score:** Increased from 2.9/5.0 to 4.7/5.0, confirming substantially higher satisfaction across all 5 dimensions (*Ease of use, Aesthetics, Speed, Clarity, Overall satisfaction*).

8.4. Qualitative Thematic Coding with Timestamps

Theme 1: Thumb ergonomics & intuitiveness of Bottom Navigation

- **P01 [0:41, 1:03]:** *"The navigation function on the old website was hidden way up in the top-left corner, forcing me to use two hands. In the new app, the bar is already under my thumb, with both icons and text, so I can operate it quickly without spending time reading."*
- **P01 [1:16]:** *"The old menu popped out and occupied the whole screen, interrupting content reading."*
- **P02 [0:37, 1:23]:** *"The old website looked like a shrunken desktop interface rather than a phone website. The new design is much clearer and easier to use with one right hand."*
- **P04 [7:11]:** *"The new app is very good for new users; when it opens, you immediately know what features the page has instead of having to search by scrolling."*
- **P05 [2:29, 3:43]:** *"The schedule is a very important function in sports. Putting it directly on the bar makes access much faster and more convenient than having to tap the hamburger menu."*
- **P06:** *"On the old website I was very confused when switching between pages and was forced to use two hands. With the new app, I could control everything with one hand after only 30 seconds of getting familiar with it."*
- **Critical perspective from P03 [0:54, 2:15]:** P03 (the only participant who preferred the hidden menu) acknowledged that the old menu was harder to reach with one hand, but personally preferred a screen without a fixed bar to save space.

Theme 2: Date-based schedule layout & 8×8 chessboard visualization

- **P01 [3:01]:** *"On the old website, the font and display style between events were too similar, making it very hard to read. The new app separates content by day in a linear way, with the number of matches shown very intuitively."*
- **P02 [3:10, 4:10]:** *"The information arrangement in the new version is compact, with lots of information but without needing to scroll far. The old version's font was very poor for skimming."*
- **P04 [2:22, 4:51]:** *"Viewing matches by day is more optimal than viewing large events, because chess viewers mainly want to know who is playing whom today. The old website's lack of a search bar is a major drawback."*
- **P05 [0:40, 1:32]:** *"The 'Go to Today' button is very useful for returning when I accidentally tap the wrong date."*
- **P06:** *"The 'Read more' button on the old website was too small to tap and it led to a news article, causing disorientation. The new app lets me view match details and has a very professional visual chessboard."*

Theme 3: Improvement contributions for future versions (Design Implications)

1. **Auto-month Threshold on Date Strip (P01 [4:12], P03 [6:54]):** Automatically update the month title on the date strip when users scroll across the boundary between two months, without waiting for users to click a specific date.
2. **Direct Date Input & Parent Tournament Header (P02 [7:05], P06):** Allow quick date/month/year entry and display the parent tournament name at the top of each day so users can follow weeks with multiple parallel

events more easily.

3. **Custom Navigation Positioning (P05 [6:01]):** Add an option to customize the position or height of the bar for users with longer thumbs or different phone-holding habits.
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9. Roles & Video Protocol

9.1. Task Assignment in Each Session

- **Facilitator — Phung Ngoc Tuan / Truong Cong Thien Phu:**
 - Welcomes participants, guides the testing steps, and gives start/stop timing commands.
 - Maintains a strictly neutral attitude, does not explain icon meanings or point at the screen unless the user asks for a hint.
 - **Observer & Data Logger — Lam Huu Khanh / Pham Chi Bao Ninh:**
 - Uses a digital stopwatch to accurately measure TCT down to milliseconds.
 - Records ERR-1, ERR-2, and ERR-3 error codes in the raw log sheet.
 - Notes Think-Aloud statements and facial reactions (surprise, confusion, satisfied smiling).
 - **Video & Tech Lead — Le Mai Hoai Bao:**
 - Sets up the camera angle: over-the-shoulder recording from a 45° side angle, fully covering the phone screen and the participant's thumb.
 - Ensures clear lighting, no screen glare, and clear participant audio recording.
 - Manages raw video files and edits evidence clips for the report slides.
-